

French Language Preparator - Specification Document

Project Overview

A web-based French language learning application that covers all CEFR levels (A1-C2) with structured lessons and certification tests modeled after **TEF Canada** and **TCF Canada** formats.

Goals

- Guide learners from absolute beginner (A1) to mastery (C2)
- Provide level-appropriate content following CEFR guidelines
- Certify proficiency at each level with TEF/TCF-style tests
- Track progress and award level certificates
- Provide user accounts (sign up / login) so each learner sees their own progress and current level
- Include pronunciation training through audio examples, phonetics, and speech practice

Competitor Analysis

Research into leading language-learning platforms (Duolingo, Babbel, Rosetta Stone, Busuu, Pimsleur, Memrise, Mondly) identified the features below. Strong, feasible features are adopted into this specification; features that are weak, outdated, or out of scope (e.g., AR/VR immersion) are noted but excluded.

Feature Comparison

Feature	Duolingo	Babbel	Rosetta Stone	Busuu	Pimsleur	Status for this App
Gamification (streaks, XP, leaderboards)	Yes	Minimal	No	No	No	Adopted
Spaced repetition (review manager)	Limited	Yes	No	Limited	Yes	Adopted

Feature	Duolingo	Babbel	Rosetta Stone	Busuu	Pimsleur	Status for this App
Grammar explanations	Minimal	Strong	None	Strong	Minimal	Adopted (strong)
Speech recognition / pronunciation	Basic	Yes	TruAccent (best)	Yes	Audio-only	Adopted (advanced)
AI conversation / roleplay	Yes (Max)	Yes	Developing	Yes	No	Adopted
AI "Explain My Answer"	Yes	No	No	No	No	Adopted
Interactive stories / dialogue scenarios	Yes	Situational	Stories	No	No	Adopted
Community corrections (native speakers)	No	No	No	Yes	No	Adopted (optional)
CEFR-aligned certification	No	No	No	Yes (McGraw-Hill)	No	Adopted (core)
Offline access	Paid	Yes	Yes	Yes	Yes	Optional (PWA)
Live classes with tutors	No	Yes	Add-on	No	No	Deferred (future)
Immersion-only (no translation)	No	No	Yes	No	No	Excluded
AR/VR lessons	No	No	No	No	No	Excluded (out of scope)
Daily goals & reminders	Yes	Limited	No	No	No	Adopted

Key Insights from Competitor Research

- **Gamification drives habit** (Duolingo) but can create "streak guilt" — implement streaks + streak-freezes with optional, non-punishing hearts system. Never block learning behind ads.
 - **Spaced repetition is proven for retention** (Pimsleur/Babbel) — vocabulary and grammar reviews must be scheduled and personalized.
 - **Strong grammar explanations** (Babbel) lead to better long-term sentence formation than pure pattern recognition.
 - **Pronunciation quality is a differentiator** (Rosetta Stone TruAccent) — our TEF/TCF focus makes speech-recognition pronunciation scoring a core feature.
 - **AI conversation unlocks real speaking practice** (Duolingo Max / Babbel AI) — simulated roleplays will be core; AI must be backed by structured grammar data to avoid confident-wrong-answers.
 - **Community corrections** (Busuu) provide human feedback AI cannot fully replace — implement as an optional native-speaker review layer.
 - **Advanced-level content thins out in most apps** (Babbel B2/C1 gap) — our C1/C2 levels are a competitive strength.
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CEFR Level Structure

Level A1 - Beginner (Découverte)

Grammar:

- Nouns: gender (masculine/feminine), number (singular/plural)
- Articles: definite (le, la, l', les), indefinite (un, une, des), partitive (du, de la, de l', des)
- Subject pronouns (je, tu, il/elle, nous, vous, ils/elles)
- Present tense of regular verbs (-er, -ir, -re)
- être (to be) and avoir (to have)
- Negation (ne...pas)
- Interrogative forms
- Demonstrative adjectives (ce, cet, cette, ces)
- Possessive adjectives (mon, ma, mes, etc.)
- Prepositions of place (à, dans, sur, sous, devant, derrière)
- Numbers (1-1000)
- Telling time
- Days, months, seasons

Vocabulary Themes:

- Greetings and introductions
- Family and relationships
- Numbers, colors, dates
- Food and drink
- Daily routine
- Weather
- Clothing
- Home and furniture
- Jobs and professions
- Time expressions

Skills:

- **Listening:** Understand slow, clear speech on familiar topics
 - **Reading:** Read simple signs, menus, schedules, short messages
 - **Writing:** Fill forms, write postcards, simple personal descriptions
 - **Speaking:** Introduce yourself, ask/answer basic questions
 - **Pronunciation:** French alphabet and letter-sound mapping, nasal vowels (on, an, in), silent letters (e muet), accent marks (é, è, ê, ç), basic intonation for statements and questions
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Level A2 - Elementary (Survie)

Grammar:

- Passé composé with avoir and être
- Imperative mood (tu, vous, nous forms)
- Object pronouns (me, te, le, la, lui, nous, vous, les, leur)
- Adjective agreement and placement
- Comparative and superlative (plus, moins, aussi)
- Demonstrative pronouns (celui, celle, ceux, celles)
- Interrogative adjectives and pronouns (quel, qui, que, quoi, où)
- Future proche (aller + infinitif)
- Conditional with si (present + present)
- Partitive articles in context
- Reflexive verbs (se lever, se coucher, etc.)
- Relative pronouns (qui, que, où)

- Adverbs of frequency and manner

Vocabulary Themes:

- Travel and transportation
- Shopping and money
- Health and body
- Hobbies and leisure
- City and directions
- Restaurant ordering
- Accommodation
- Technology basics
- Emotions and feelings
- Celebrations and traditions

Skills:

- **Listening:** Understand short conversations on familiar topics
 - **Reading:** Understand personal letters, short articles, advertisements
 - **Writing:** Write short personal letters, describe experiences, make simple lists
 - **Speaking:** Describe experiences, express opinions simply, handle travel situations
 - **Pronunciation:** Common phoneme contrasts (é/è, u/ou), pronunciation of final consonants, basic liaison (les amis, nous avons), rhythmic group and stress on the final syllable, intonation in questions
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Level B1 - Intermediate (Seuil)

Grammar:

- Passé composé vs Imparfait (distinction and usage)
- Plus-que-parfait (pluperfect)
- Passé simple (literary past)
- Future simple
- Conditional present
- Passive voice (être + past participle)
- Causative (faire + infinitive)
- Relative pronouns (dont, lequel, auquel, duquel)
- Infinitive constructions (pour + infinitive, sans + infinitive, avant de, après de)
- Subjonctif présent (introduction)

- Discourse markers (cependant, toutefois, néanmoins, en revanche)
- Indirect speech (reported speech)
- Comparisons of equality and inequality
- Time expressions with prepositions

Vocabulary Themes:

- Work and career
- Education and studies
- Media and news
- Environment and nature
- Social issues
- Culture and arts
- Sports and fitness
- Relationships and emotions
- Politics and society
- Science and technology

Skills:

- **Listening:** Understand main points of clear standard speech on familiar matters
- **Reading:** Understand texts that consist mainly of high-frequency language
- **Writing:** Write connected text on familiar topics, describe experiences and events
- **Speaking:** Express opinions, give reasons, narrate stories
- **Pronunciation:** Full liaison and elision rules, intonation for expressing emotions and attitudes, rhythm and flow in connected speech, pronunciation of literary/formal French versus everyday French

Level B2 - Upper Intermediate (Avancé)

Grammar:

- Subjonctif présent (full usage)
- Subjonctif imparfait (literary)
- Conditionnel passé (would have done)
- Passé composé vs Plus-que-parfait vs Conditionnel passé
- Subordonnant relatives (dont, lequel, etc.) - advanced
- Cleft sentences (c'est...qui/que)
- Emphatic structures

- Advanced passive voice
- Inversion in formal register
- Hypothetical sentences with conditional
- Reported speech - advanced
- Advanced conjunctions and connectors
- Relative clauses - reduced forms
- Past conditional
- Sequence of tenses

Vocabulary Themes:

- Abstract concepts (justice, freedom, equality)
- Business and economics
- Law and legal system
- Philosophy and ethics
- International relations
- Literature and film analysis
- Scientific discourse
- Nuanced emotions and states
- Technical vocabulary (chosen domain)
- Idiomatic expressions

Skills:

- **Listening:** Understand extended speech and follow complex arguments
- **Reading:** Understand articles and reports on contemporary issues
- **Writing:** Write clear, detailed text on a wide range of subjects
- **Speaking:** Present clear arguments, defend opinions, interact with fluency
- **Pronunciation:** Advanced liaison and enchaînement, nuanced intonation for rhetorical effect, conscious control of pitch and tempo, awareness of regional and francophone accents (Québec, Belgique, Suisse, Afrique)

Level C1 - Advanced (Autonomie)

Grammar:

- Subjonctif passé and imparfait (advanced usage)
- Advanced conditional structures

- Complex sentence structures
- Nuanced passive and middle voice
- Literary tenses and registers
- Stylistic inversion
- Advanced discourse organization
- Pleonastic structures
- Advanced pronoun usage
- Complex temporal and causal relationships
- Register shifts (formal/informal)
- Advanced prepositional phrases
- Nominalization

Vocabulary Themes:

- Academic and research language
- Literary and artistic criticism
- Diplomatic and political discourse
- Nuanced abstract reasoning
- Specialized professional vocabulary
- Cultural references and allusions
- Advanced idiomatic expressions
- Register-specific vocabulary
- Technical writing
- Persuasive language

Skills:

- **Listening:** Understand extended speech even when it is not clearly structured
- **Reading:** Understand long and complex factual and literary texts
- **Writing:** Write clear, well-structured, detailed text on complex subjects
- **Speaking:** Express ideas fluently, use language flexibly for social and professional purposes
- **Pronunciation:** Perfect intonation patterns, mastering prosody for emphasis and persuasion, adjusting pronunciation to register (formal, informal, academic), flawless rhythm in fast speech

Level C2 - Mastery (Maîtrise)

Grammar:

- Full mastery of all tenses and moods
- Subtle nuances in tense usage
- Advanced stylistic devices
- Perfect register control
- Complex syntactic structures
- Literary and rhetorical devices
- Advanced concordance of tenses
- Masterful use of connectors
- Sophisticated pronoun reference
- Advanced negation structures
- Pleonasm and emphasis techniques

Vocabulary Themes:

- Near-native vocabulary range
- Specialized terminology (multiple fields)
- Advanced wordplay and humor
- Nuanced synonyms and near-synonyms
- Etymology and word formation
- Register-appropriate vocabulary
- Cultural and historical references
- Advanced collocations
- Figurative language mastery
- Technical jargon as needed

Skills:

- **Listening:** Understand virtually everything heard with ease
 - **Reading:** Read and assimilate complex texts of abstract nature
 - **Writing:** Write any type of text with appropriate style and structure
 - **Speaking:** Express yourself spontaneously, precisely, and fluently
 - **Pronunciation:** Native-like prosody and intonation, mastery of rhythm, pauses and stress for rhetorical effect, unrestricted use of all registers, flawless pronunciation in any context
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Certification Tests (TEF Canada / TCF Canada Format)

Test Structure Overview

Each level certification test follows the official TEF Canada and TCF Canada format with four sections:

Section	Duration	Questions	Weight
Compréhension orale (Listening)	30-40 min	30-40 questions	25%
Compréhension écrite (Reading)	45-60 min	30-40 questions	25%
Expression écrite (Writing)	45-60 min	2-3 tasks	25%
Expression orale (Speaking)	12-15 min	2-3 tasks	25%

A1 Level Test

Listening (Compréhension orale)

- 30 questions
- Identify specific information in short, slow recordings
- Match speakers to topics
- Understand simple instructions
- Recognize numbers, dates, times
- **Duration:** 30 minutes

Reading (Compréhension écrite)

- 30 questions
- Understand simple written messages
- Find information in advertisements, notices
- Complete simple forms
- Read short, simple texts
- **Duration:** 45 minutes

Writing (Expression écrite)

- 2 tasks:
 1. Complete a form (personal information)

2. Write a short message (postcard, email - 30-40 words)

- **Duration:** 45 minutes

Speaking (Expression orale)

- 2 tasks:
 1. Give personal information (interview simulation)
 2. Ask and answer simple questions
- **Duration:** 12 minutes

Passing Score: 150-200 points (out of 300) **Certificate:** A1 Level Certified

A2 Level Test

Listening (Compréhension orale)

- 35 questions
- Understand short conversations on familiar topics
- Follow simple directions
- Identify main idea in short recordings
- Understand announcements
- **Duration:** 35 minutes

Reading (Compréhension écrite)

- 35 questions
- Understand personal correspondence
- Read short newspaper articles
- Follow simple instructions
- Understand timetables and brochures
- **Duration:** 50 minutes

Writing (Expression écrite)

- 2 tasks:
 1. Write a personal message (email, postcard - 60-80 words)
 2. Describe a photo or situation briefly
- **Duration:** 50 minutes

Speaking (Expression orale)

- 3 tasks:
 1. Introduce yourself and give personal information
 2. Describe a photo
 3. Handle a transactional situation (ordering, asking directions)
- **Duration:** 13 minutes

Passing Score: 200-250 points **Certificate:** A2 Level Certified

B1 Level Test

Listening (Compréhension orale)

- 35 questions
- Understand the main points of standard speech
- Follow a conversation between native speakers
- Understand radio broadcasts on familiar topics
- Identify speaker's attitude/opinion
- **Duration:** 35 minutes

Reading (Compréhension écrite)

- 35 questions
- Understand articles on contemporary issues
- Read personal and professional correspondence
- Understand factual texts
- Identify main points and supporting details
- **Duration:** 55 minutes

Writing (Expression écrite)

- 2 tasks:
 1. Write a personal letter or email (120-150 words)
 2. Narrate an experience or tell a story
- **Duration:** 55 minutes

Speaking (Expression orale)

- 3 tasks:
 1. Give a structured presentation on a familiar topic
 2. Narrate an experience
 3. Defend an opinion with reasons
- **Duration:** 14 minutes

Passing Score: 250-300 points **Certificate:** B1 Level Certified

B2 Level Test

Listening (Compréhension orale)

- 40 questions
- Understand extended speech and lectures
- Follow complex arguments
- Understand TV programs, documentaries
- Identify implicit meaning
- **Duration:** 40 minutes

Reading (Compréhension écrite)

- 40 questions
- Understand complex articles on abstract topics
- Read contemporary literary texts
- Understand specialist articles outside field
- Identify author's purpose and tone
- **Duration:** 60 minutes

Writing (Expression écrite)

- 2 tasks:
 1. Write an essay or article (200-250 words)
 2. Write a formal letter expressing opinion
- **Duration:** 60 minutes

Speaking (Expression orale)

- 3 tasks:
 1. Present and defend a viewpoint
 2. Discuss abstract topics
 3. Role-play a professional/social situation
- **Duration:** 15 minutes

Passing Score: 300-375 points **Certificate:** B2 Level Certified

C1 Level Test

Listening (Compréhension orale)

- 40 questions
- Understand extended speech even when not clearly structured
- Understand TV shows, films in standard dialect
- Follow complex technical discussions
- Understand implicit meaning and attitude
- **Duration:** 40 minutes

Reading (Compréhension écrite)

- 40 questions
- Understand long and complex texts
- Read academic and literary works
- Understand specialized articles
- Identify fine points of argument
- **Duration:** 60 minutes

Writing (Expression écrite)

- 2 tasks:
 1. Write a detailed text on complex subjects (250-300 words)
 2. Write an academic or professional document
- **Duration:** 60 minutes

Speaking (Expression orale)

- 3 tasks:
 1. Give a formal presentation
 2. Participate in a debate
 3. Handle complex professional situations
- **Duration:** 15 minutes

Passing Score: 375-450 points **Certificate:** C1 Level Certified

C2 Level Test

Listening (Compréhension orale)

- 40 questions
- Understand virtually everything heard with ease
- Follow fast-paced native speech
- Understand humor, irony, cultural references
- Understand multiple speakers simultaneously
- **Duration:** 40 minutes

Reading (Compréhension écrite)

- 40 questions
- Read and understand any type of text
- Appreciate literary and stylistic nuances
- Understand specialized, technical, or abstract texts
- Identify subtle implications
- **Duration:** 60 minutes

Writing (Expression écrite)

- 2 tasks:
 1. Write a complex, well-structured text (300+ words)
 2. Produce a professional or academic document with precision
- **Duration:** 60 minutes

Speaking (Expression orale)

- 3 tasks:
 1. Give an extended presentation on any topic
 2. Participate fluently in spontaneous conversation
 3. Handle any social, academic, or professional situation
- **Duration:** 15 minutes

Passing Score: 450-540 points **Certificate:** C2 Level Certified (Mastery)

Progression Rules

1. **Placement Check:** New users may take a quick placement quiz to start at their actual level; otherwise they begin at A1
 2. **Sequential Completion:** Must complete Level N before attempting Level N+1 test
 3. **Minimum Study Time:** Recommended minimum study hours per level:
 - A1: 60-80 hours
 - A2: 60-80 hours
 - B1: 100-120 hours
 - B2: 120-150 hours
 - C1: 150-180 hours
 - C2: 180-200+ hours
 4. **Practice Tests:** Unlimited practice tests available before certification test
 5. **Retake Policy:** Can retake certification test after 7 days
 6. **Certificate Generation:** Verifiable digital certificate generated upon passing with score, skill breakdown, and date
 7. **Daily Practice:** Configurable daily goals (lessons/XP) with reminders to build consistent habits
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Technical Requirements

- **Framework:** Next.js (App Router) with composable React components
- **Styling:** Tailwind CSS / CSS Modules, mobile-first responsive design
- **Authentication:** JWT-based auth with OAuth providers (Google, GitHub) and email/password (NextAuth/Auth.js)
- **Database:** PostgreSQL with Prisma ORM (users, progress, certificates, gamification, SRS review data)

- **Rendering:** React Server Components for content pages, Client Components for interactive exercises and tests
 - **Audio:** HTML5 Audio API / Web Speech API (SpeechSynthesis for pronunciation examples, SpeechRecognition for pronunciation feedback)
 - **AI Features:** Server-side AI API integration (roleplay conversations, "Explain My Answer", adaptive difficulty, AI tutor Q&A)
 - **Real-time:** API Routes / Server Actions for test submissions and progress updates
 - **Background Jobs:** Cron for daily goal reminders (email/in-app notifications) and SRS scheduling
 - **Spaced Repetition:** SM-2/Leitner scheduling algorithm implementable with plain SQL (no external dependency)
 - **Certificates:** PDF generation (server-side) with unique verification codes
 - **Offline:** PWA / Service Worker for offline access (optional)
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UI/UX Structure

1. **Landing Page:** Welcome page with Sign Up / Login buttons
2. **Sign Up Page:** Create account (name, email, password) or OAuth
3. **Login Page:** Email/password or OAuth login
4. **Placement Check:** Quick quiz to determine starting CEFR level
5. **Dashboard (Home):** Personalized level selection, current level, progress summary, daily goal progress, streak counter, review queue count
6. **Level Pages:** Lessons organized by skill (Grammar, Vocabulary, Listening, Reading, Writing, Speaking, Pronunciation)
7. **Pronunciation Pages:** Phoneme examples with audio playback, IPA charts, tongue/mouth positioning diagrams, and speech-recognition practice
8. **Practice Pages:** Interactive exercises for each skill
9. **Review Page (SRS):** Spaced-repetition flashcard queue for vocabulary/grammar reviews
10. **Stories Page:** Interactive dialogue scenarios with comprehension questions
11. **AI Conversation Page:** AI roleplay chat, "Explain My Answer" explanations
12. **Community Page:** Written/speaking submissions for native-speaker corrections
13. **Gamification:** Streak display, XP counter, badges, leaderboards (top bar / dashboard widgets)
14. **Test Pages:** Certification test interface with timer
15. **Results Page:** Score breakdown, certificate display
16. **Profile Page:** User info, personal progress tracking, earned certificates, settings

User Accounts & Progress Tracking

- **Sign Up:** Create account with email + password, or OAuth (Google/GitHub). Fields: name, email, target CEFR level, native language.
- **Login:** Authenticate returning users and restore their personal dashboard.
- **Progress Tracking (per user):**
 - Current level and unlocked levels
 - Completed lessons and exercises for each skill
 - Practice test scores history
 - Certification test attempts and results
 - Time spent per level
 - Pronunciation accuracy scores per phoneme/lesson
- **State Synchronization:** Progress is saved to the database after each lesson/exercise/test; the dashboard reflects real-time updates across devices.

Pronunciation Module

- **Phonetics:** Each level introduces IPA notation and key French sounds (voyelles orales, voyelles nasales, consonnes, semi-voyelles).
- **Audio Examples:** Click-to-play native audio for every word, phrase, phoneme, and minimal pair.
- **Text-to-Speech:** Built-in TTS to hear any French text spoken at a chosen speed (slow → normal → fast).
- **Speech Recognition:** Learners record themselves and get instant feedback (accuracy score, closest-match feedback, replay of their recording vs. native sample).
- **Minimal Pairs Drills:** Contrast tricky sounds (u/ou, é/è, an/on/in, b/v, s/z, p/b) with listening + speaking exercises.
- **Liaison & Elision:** Structured lessons teaching when to link words (les amis, nous #avons) with interactive examples.
- **Intonation Training:** Question vs. statement vs. exclamation curves, emotion expression, with visual pitch graphs.
- **Pronunciation Test:** Part of the certification test - evaluated with speech recognition scoring for each CEFR level.

Premium Features (from Competitor Analysis)

Features adopted because they are proven to improve learning outcomes and engagement:

1. Gamification Engine

- **Streaks:** Daily practice streak counter + "streak freeze" items so accidental misses do not cause streak loss.
- **XP & Levels:** Points earned per lesson/exercise; user level-up system (no ads, no pay-to-progress).
- **Badges/Achievements:** Milestone badges (e.g., "First 100 Words", "7-Day Streak", "A1 Certified", "Perfect Pronunciation").
- **Daily Goals & Reminders:** User-configurable daily XP/lesson targets with in-app and email reminders.
- **Leaderboards:** Optional weekly leagues / friend leaderboards (privacy-controlled; can be hidden).
- **Hearts/Lives (optional):** Soft-limit mistakes during certification tests only; practice mode is unlimited.

2. Spaced Repetition System (SRS)

- **Review Manager:** Automatically schedules vocabulary and grammar reviews at optimal intervals (SM-2 / Leitner algorithm).
- **Flashcards:** Word/Phrase -> translation -> audio playback -> speech-recognition self-check.
- **Integration:** New words from lessons, stories, AI chats, and mistakes automatically enter the review queue.
- **Daily Review Session:** Short, personalized review queue each day based on due items.

3. Interactive Stories & Real-Life Scenarios

- **Stories:** Short interactive dialogues with native audio (like Duolingo Stories): listen -> answer comprehension questions -> vocabulary extraction.
- **Situational Lessons:** Real scenarios (ordering food, checking into a hotel, job interview, TEF/TCF test simulation) with branching dialogues.
- **Fill-in & Roleplay:** Some stories let the learner take a role and respond via speech recognition.

4. AI Conversation Partner & Helpers

- **AI Roleplay:** Simulated conversations with AI characters across realistic scenarios, with the AI adapting to the learner's level.
- **AI Video-Style Calls:** (Optional, advanced) conversation practice with a simulated native speaker, evaluated for grammar and pronunciation.
- **"Explain My Answer":** After any mistake, one tap gives a clear grammar/vocabulary explanation so learners understand *why* an answer is wrong.
- **AI Tutor Q&A:** Ask grammar questions in natural language; answers are grounded in the structured grammar database to prevent confident-wrong answers.

5. Adaptive Learning Path

- **Personalized Difficulty:** AI adjusts question difficulty based on the learner's mastery of each skill and past mistakes.
- **Weak-Skill Targeting:** The app recommends lessons/practice for underperforming skills (e.g., low pronunciation or listening scores).
- **Placement Check:** New users can take a quick placement quiz to start at their actual level instead of A1.

6. Community Corrections (Native Speaker Review)

- **Written Submissions:** Learners submit short writing exercises; native speakers (or AI-assisted review) correct and annotate them.
- **Spoken Submissions:** Learners record themselves and receive pronunciation feedback from the community.
- **Review Exchange:** Optional reciprocal feature where learners who are native French speakers review others' work and earn badges.
- **Fallback:** AI-assisted auto-feedback when community review is slow.

7. Enhanced Certificates

- **Verification System:** Every certificate has a unique verification code that can be checked by third parties (employers, universities).
- **Shareable:** Download as PDF, shareable link, and LinkedIn-ready graphic with score + date + CEFR level.
- **Progress Record:** Certificate includes per-skill breakdown (listening, reading, writing, speaking, pronunciation, grammar, vocabulary).

8. Cultural Insights

- **Culture Notes:** Cultural context embedded in lessons (France, Québec, Belgique, Suisse, Maghreb, Afrique francophone).
 - **Francophone Spotlight:** Every level includes authentic short texts/audio from different francophone regions.
 - **Etiquette & Pragmatics:** Formality (tu/vous), gestures, and social norms relevant to each level.
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Content Sources

Primary source material: `Complete French All-in-One` (**Practice Makes Perfect, McGraw-Hill**) — a CEFR-aligned comprehensive French grammar and exercise book included in this project. The curriculum extracts grammar topics, vocabulary, dialogues, and exercise patterns from this book for all six levels.

Supplementary sources:

- Official TEF Canada / TCF Canada sample tests and topic guides (for test structure, question types, and scoring).
- Native-speaker recordings / TTS for all audio content.
- Open francophone corpora and news sources for authentic B1-C2 reading/listening materials.
- IPA (International Phonetic Alphabet) charts for the pronunciation module.

Content mapping: Every grammar topic in the CEFR Level Structure above is mapped to specific chapters/exercises from `Complete French All-in-One`, ensuring pedagogical accuracy and depth at every level.

Data Structure (Content & User Storage)

Content (lesson data):

```
{
  "level": "A1",
  "topics": [
    {
      "id": "a1-grammar-01",
      "title": "Nouns and Articles",
      "lessons": [...],
      "exercises": []
    }
  ],
  "pronunciation": {
    "phonemes": [{ "symbol": "ã", "examples": ["vent", "enfant"] }],
    "minimalPairs": [{ "a": "dessus", "b": "dessous", "audio": true }],
    "lessons": [...]
  },
  "test": {
    "listening": [...],
```

```
"reading": [...],
"writing": [...],
"speaking": [...],
"pronunciation": [...]
}
}
```

User / Progress (database):

```
{
  "user": {
    "id": "uuid",
    "name": "John Doe",
    "email": "john@example.com",
    "authProvider": "email | google | github",
    "currentLevel": "B1",
    "unlockedLevels": ["A1", "A2", "B1"],
    "targetLevel": "C1",
    "nativeLanguage": "English",
    "createdAt": "ISO8601"
  },
  "progress": {
    "userId": "uuid",
    "completedLessons": ["a1-grammar-01", "a1-vocab-02"],
    "skillScores": {
      "grammar": 85,
      "vocabulary": 90,
      "listening": 78,
      "reading": 82,
      "writing": 74,
      "speaking": 70,
      "pronunciation": 68
    },
    "practiceTests": [{ "level": "A1", "date": "ISO8601", "score": 200 }],
    "certifications": [
      { "level": "A1", "date": "ISO8601", "score": 210, "certificateId": "UUID" }
    ],
    "pronunciationScores": [{ "phoneme": "ā", "accuracy": 87, "date": "ISO8601" }]
  },
  "gamification": {
```

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"userId": "uuid",
"xp": 12450,
"streakCurrent": 23,
"streakLongest": 45,
"dailyGoalXp": 50,
"dailyGoalMet": ["2026-09-01", "2026-09-02"],
"badges": ["first-100-words", "7-day-streak", "a1-certified"],
"league": "Bronze | Silver | Gold | Diamond"
},
"srsReviews": {
  "userId": "uuid",
  "queue": [
    { "item": "a1-vocab-chair", "dueDate": "ISO8601", "interval": 4, "repetitions": 2, "easeFactor"
  ]
},
"aiHistory": {
  "userId": "uuid",
  "roleplaySessions": [{ "date": "ISO8601", "scenario": "hotel-checkin", "score": 82 }],
  "explanations": [{ "date": "ISO8601", "question": "...", "category": "grammar" }]
},
"community": {
  "userId": "uuid",
  "writtenSubmissions": [{ "id": "uuid", "date": "ISO8601", "status": "pending | corrected" }],
  "spokenSubmissions": [{ "id": "uuid", "date": "ISO8601", "feedbackRating": 4 }],
  "reviewsGiven": 12
},
"certificate": {
  "id": "UUID",
  "userId": "uuid",
  "level": "A1",
  "score": 210,
  "skillBreakdown": { "listening": 82, "reading": 85, "writing": 78, "speaking": 70, "pronunciation"
  "issuedAt": "ISO8601",
  "verificationCode": "string"
}
}
```

Exercise Types Taxonomy

The lesson engine supports these exercise types across all skills and levels:

Type	Skill	Description	Example
multiple_choice	Grammar, Vocab, Reading, Listening	Single or multiple correct answers	"Choose the correct article: ___ livre"
fill_in_blank	Grammar, Vocab, Writing	Type missing word/phrase	"Je ___ (manger) une pomme"
translation	Grammar, Vocab, Writing	Translate sentence EN↔FR	"Translate: 'I am eating'"
drag_drop	Grammar, Vocab	Reorder words / match pairs	"Put words in order: [mange / je / pomme / une]"
conjugation_drill	Grammar	Conjugate verb in given tense/pronoun	"Conjugate 'manger' - nous, présent"
listening_comprehension	Listening	Audio → question (MC, true/false, gap-fill)	"Listen and choose the correct picture"
dictation	Listening, Writing	Audio → type exactly what you hear	"Écrivez ce que vous entendez"
pronunciation	Speaking, Pronunciation	Record word/phrase → speech recognition score	"Say: 'bonjour' — score: 92/100"
speaking_response	Speaking	Prompt → record spoken answer (AI/human evaluated)	"Describe your daily routine"
reading_comprehension	Reading	Text → questions (MC, matching, true/false)	"Read the email and answer questions"
image_match	Vocab, Listening	Match word/audio to image	"Match audio to correct food image"
sentence_builder	Grammar, Writing	Build sentence from word bank	"Build: 'I [like] [eating] [apples]'"
minimal_pair_drill	Pronunciation	Contrast two similar sounds	"Click the word you hear: 'dessus' /

Type	Skill	Description	Example
			'dessous'"
liaison_practice	Pronunciation	Identify/produce liaison in sentences	"Mark where liaison occurs: 'les amis'"
roleplay_turn	Speaking, AI	AI character speaks → learner responds	"AI: 'Bonjour, une table pour deux?' → You: ..."

Component mapping: Each type maps to a reusable React component (`<MultipleChoice />` , `<FillInBlank />` , etc.) with props for content, validation, and feedback.

Test Question Taxonomy (TEF/TCF Format)

Compréhension orale (Listening)

Question Type	Format	Levels	Example
audio_mc_single	Audio + 3-4 options, one correct	All	"Que fait l'homme ?"
audio_mc_multiple	Audio + select all that apply	B1+	"Quels sujets sont abordés ?"
audio_true_false	Audio + statement → Vrai/Faux	A1-B2	"Le train part à 10h. Vrai/Faux"
audio_gap_fill	Audio with gaps → type missing words	B1+	"Le concert commence à ____"
audio_image_match	Audio → choose matching image	A1-A2	"Select the picture of 'la gare'"
audio_ordering	Audio segments → put in order	B2+	"Reconstituez le dialogue"

Compréhension écrite (Reading)

Question Type	Format	Levels	Example
text_mc_single	Text + 3-4 options	All	"Quel est le thème principal ?"
text_mc_multiple	Text + select all correct	B1+	"Quelles affirmations sont vraies ?"
text_true_false	Text + statements → Vrai/Faux	A1-B2	"L'auteur est médecin. Vrai/Faux"

Question Type	Format	Levels	Example
text_gap_fill	Text with blanks → choose word	A2+	"Il ____ (mange/manges/mangeons) une pomme"
text_matching	Match headings/people to paragraphs	B1+	"Match each person to their opinion"
text_ordering	Sentences → logical order	B2+	"Remettez le paragraphe dans l'ordre"
text_short_answer	Text → write short answer	C1+	"En deux phrases, résumez l'argument"

Expression écrite (Writing)

Task Type	Format	Levels	Word Count
form_completion	Fill official form	A1-A2	N/A
short_message	Postcard, email, SMS	A1-B1	30-80 words
describe_photo	Describe image in sentences	A2-B1	40-60 words
personal_letter	Informal letter/email	B1-B2	120-150 words
opinion_essay	Structured essay with arguments	B2-C2	200-300+ words
formal_letter	Complaint, application, request	B2-C2	150-250 words
report_summary	Summarize data/text	C1-C2	250-300 words

Expression orale (Speaking)

Task Type	Format	Levels	Duration
guided_interview	Examiner asks personal questions	All	2-3 min
picture_description	Describe photo	A2-C1	1-2 min
roleplay_transactional	Simulated scenario (shop, hotel)	A2-B2	2-3 min
presentation	Prepared talk on topic	B1-C2	2-3 min
debate_defend	Defend viewpoint with arguments	B2-C2	3-4 min
complex_situation	Handle professional/abstract scenario	C1-C2	3-4 min

Scoring: Each question/task has a rubric mapped to CEFR descriptors. Auto-scored (MC, gap-fill) vs. human/AI-scored (writing, speaking).

Pronunciation Scoring Rubric

Speech recognition returns a composite score (0-100) weighted from sub-scores:

Sub-score	Weight	Description	Measurement
Phoneme Accuracy	50%	Correct articulation of vowels, consonants, nasals	Forced alignment + phoneme-level confidence
Prosody	25%	Rhythm, stress, intonation patterns	Pitch contour comparison, syllable duration ratios
Fluency	15%	Speech rate, pausing, hesitation markers	Words/min, pause frequency/duration, fillers
Completeness	10%	All required words spoken, no omissions	Word coverage vs. reference transcript

CEFR Band Mapping:

Composite	CEFR Band	Description
90-100	C2	Native-like, effortless
80-89	C1	Clear, natural, minor L1 traces
70-79	B2	Good control, occasional errors
60-69	B1	Understandable, noticeable accent
50-59	A2	Frequent errors, effort required
<50	A1	Largely unintelligible

Test Integration: Pronunciation section score = average of 5-10 pronunciation tasks. Contributes 25% of Expression orale score (or separate Pronunciation certificate component).

AI Integration Spec

Feature	Provider	Model	Token Budget	Fallback
AI Roleplay	OpenAI	gpt-4o-mini	2k input / 1k output per turn	Rule-based dialogue tree
Explain My Answer	OpenAI	gpt-4o-mini	1k input / 500 output	Canned grammar explanations from DB
AI Tutor Q&A	OpenAI	gpt-4o	3k input / 1k output	Search grammar DB + template reply
Adaptive Difficulty	Local	Custom ML (scikit-learn/TF.js)	N/A	Heuristic: mastery < 60% → easier
Content Generation (dev)	OpenAI	gpt-4o	4k input / 2k output	Human-authored only

Cost Estimates (per 1000 active users/day):

- Roleplay: ~50 turns/user/day × \$0.00015 = \$7.50/day
- Explain My Answer: ~20 requests/user/day × \$0.0001 = \$2.00/day
- Tutor Q&A: ~5 requests/user/day × \$0.0005 = \$2.50/day
- **Total: ~\$12/day / \$360/month** (at gpt-4o-mini pricing)

Safety: All prompts include system instructions to refuse non-language content, avoid confident-wrong answers, and cite grammar rules from structured DB.

Latency Targets: Roleplay < 800ms, Explain < 500ms, Tutor < 1200ms (p95).

Prisma Schema (PostgreSQL)

```
generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}
```

```

}

model User {
  id          String    @id @default(uuid())
  name        String
  email       String    @unique
  passwordHash String?
  authProvider AuthProvider @default(EMAIL)
  currentLevel CEFRLevel @default(A1)
  targetLevel  CEFRLevel @default(C1)
  nativeLanguage String  @default("en")
  createdAt    DateTime  @default(now())
  updatedAt    DateTime  @updatedAt

  progress      Progress?
  gamification  Gamification?
  srsReviews    SrsReview[]
  aiHistory     AiHistory?
  community     Community?
  certificates  Certificate[]

  @@index([email])
  @@index([currentLevel])
}

enum AuthProvider {
  EMAIL
  GOOGLE
  GITHUB
}

enum CEFRLevel {
  A1
  A2
  B1
  B2
  C1
  C2
}

model Progress {

```

```

id          String  @id @default(uuid())
userId      String  @unique
user        User    @relation(fields: [userId], references: [id], onDelete: Cascade)
completedLessons String[] @default([])
skillScores Json    @default("{}")
practiceTests Json    @default("[]")
certifications Json    @default("[]")
pronunciationScores Json @default("[]")
updatedAt   DateTime @updatedAt
}

```

```

model Gamification {
  id          String  @id @default(uuid())
  userId      String  @unique
  user        User    @relation(fields: [userId], references: [id], onDelete: Cascade)
  xp          Int     @default(0)
  streakCurrent Int    @default(0)
  streakLongest Int    @default(0)
  dailyGoalXp Int     @default(50)
  dailyGoalMet DateTime[] @default([])
  badges      String[] @default([])
  league      League   @default(BRONZE)
  updatedAt   DateTime @updatedAt
}

```

```

enum League {
  BRONZE
  SILVER
  GOLD
  DIAMOND
}

```

```

model SrsReview {
  id          String  @id @default(uuid())
  userId      String
  user        User    @relation(fields: [userId], references: [id], onDelete: Cascade)
  itemId      String  // e.g., "a1-vocab-chair"
  itemType    SrsItemType
  dueDate     DateTime
  interval    Int     @default(1)
  repetitions Int     @default(0)
}

```

```

easeFactor    Float    @default(2.5)
lastReviewed  DateTime?
createdAt     DateTime @default(now())
updatedAt     DateTime @updatedAt

@@unique([userId, itemId])
@@index([userId, dueDate])
}

enum SrsItemType {
    VOCAB
    GRAMMAR
    PHRASE
    PHONEME
}

model AiHistory {
    id          String    @id @default(uuid())
    userId      String    @unique
    user        User      @relation(fields: [userId], references: [id], onDelete: Cascade)
    roleplaySessions Json  @default("[]")
    explanations Json      @default("[]")
    updatedAt   DateTime @updatedAt
}

model Community {
    id          String    @id @default(uuid())
    userId      String    @unique
    user        User      @relation(fields: [userId], references: [id], onDelete: Cascade)
    writtenSubmissions Json  @default("[]")
    spokenSubmissions Json  @default("[]")
    reviewsGiven Int       @default(0)
    reputation   Int       @default(0)
    updatedAt   DateTime @updatedAt
}

model Certificate {
    id          String    @id @default(uuid())
    userId      String
    user        User      @relation(fields: [userId], references: [id], onDelete: Cascade)
    level       CEFRLevel

```

```

score          Int
skillBreakdown Json
issuedAt       DateTime @default(now())
verificationCode String  @unique

@@index([userId])
@@index([verificationCode])
}

// Content models (seeded from Complete French All-in-One)
model LevelContent {
  id          String  @id @default(uuid())
  level       CEFRLevel @unique
  grammar     Json    // topics, lessons, exercises
  vocabulary  Json
  pronunciation Json
  test        Json
  stories     Json    @default("[]")
  scenarios   Json    @default("[]")
  culturalNotes Json  @default("[]")
  bookMapping Json    @default("{}") // chapter/exercise refs
  updatedAt   DateTime @updatedAt
}

```

API Routes / Server Actions

Authentication

- `POST /api/auth/register` — email/password signup
- `POST /api/auth/login` — email/password login
- `GET /api/auth/session` — get current session
- `POST /api/auth/logout` — destroy session
- `GET/POST /api/auth/oauth/[provider]` — NextAuth OAuth flow

User & Progress

- `GET /api/user/profile` — current user profile
- `PATCH /api/user/profile` — update name, target level, native language, daily goal

- `GET /api/user/progress` — full progress object (skillScores, completedLessons, etc.)
- `POST /api/user/progress/lesson` — mark lesson complete, update skillScores
- `GET /api/user/stats` — dashboard stats (streak, xp, review count, time spent)

Lessons & Content

- `GET /api/levels/[level]/content` — full level content (grammar, vocab, pronunciation, stories)
- `GET /api/levels/[level]/lessons/[topicId]` — single lesson with exercises
- `GET /api/levels/[level]/pronunciation` — phonemes, minimal pairs, audio URLs
- `GET /api/levels/[level]/stories` — interactive stories list
- `GET /api/levels/[level]/stories/[storyId]` — story content + comprehension questions
- `GET /api/levels/[level]/scenarios` — situational dialogues

Exercises & Practice

- `POST /api/exercises/submit` — submit exercise answer → returns score + feedback
- `POST /api/exercises/pronunciation` — submit audio blob → returns pronunciation score breakdown
- `GET /api/exercises/adaptive` — get next exercise based on weak skills

Spaced Repetition (SRS)

- `GET /api/srs/queue` — today's due review items
- `POST /api/srs/review` — submit review result (again/hard/good/easy) → updates interval
- `GET /api/srs/stats` — review stats (due, learned, retention rate)

AI Features

- `POST /api/ai/roleplay` — send user message → AI response + evaluation
- `POST /api/ai/explain` — send question/exercise context → explanation
- `POST /api/ai/tutor` — send grammar question → grounded answer
- `POST /api/ai/adaptive` — send skill mastery data → difficulty recommendation

Tests & Certification

- `GET /api/tests/[level]/practice` — generate practice test
- `POST /api/tests/[level]/practice/submit` — submit practice test → score + breakdown
- `POST /api/tests/[level]/certification/start` — start timed certification test
- `POST /api/tests/[level]/certification/submit` — submit certification test
- `GET /api/tests/[level]/certification/result` — get result + certificate if passed

- `GET /api/certificates/[certificateId]` — public certificate verification
- `GET /api/certificates/[certificateId]/pdf` — download certificate PDF
- `GET /api/certificates/[certificateId]/share` — shareable image/link

Community

- `POST /api/community/submit/written` — submit writing for correction
- `POST /api/community/submit/spoken` — submit audio for pronunciation feedback
- `GET /api/community/queue` — get submissions to review (for native speakers)
- `POST /api/community/review` — submit correction/feedback
- `GET /api/community/my-submissions` — user's submissions with status

Gamification

- `GET /api/gamification/stats` — XP, streak, badges, league
- `POST /api/gamification/daily-goal` — mark daily goal met
- `GET /api/gamification/leaderboard` — weekly league standings
- `GET /api/gamification/badges` — all badges with unlock criteria

Admin / Content (protected)

- `GET /api/admin/content/levels` — list all level content
- `POST /api/admin/content/levels/[level]` — upsert level content
- `POST /api/admin/content/import` — import from Complete French All-in-One mapping
- `GET /api/admin/users` — user management
- `GET /api/admin/analytics` — platform analytics (engagement, completion, retention)

Success Criteria

- ☐ All 6 CEFR levels implemented with complete content
- ☐ Sign up / login with email-password and OAuth providers
- ☐ Personalized dashboard showing each user's own progress and current level
- ☐ Progress saved per user across devices (database-backed, not just LocalStorage)
- ☐ TEF/TCF-style tests for each level
- ☐ Pronunciation module per level with phonemes, minimal pairs, and audio examples
- ☐ Speech-recognition-based pronunciation practice and scoring
- ☐ Pronunciation included in certification tests

- ☐ Score calculation and verifiable certificate generation (unique codes, shareable)
- ☐ Progress tracking across levels
- ☐ Gamification: streaks, XP, badges, daily goals, optional leaderboards
- ☐ Spaced repetition review manager for vocabulary and grammar retention
- ☐ Interactive stories/roleplay scenarios with native audio
- ☐ AI conversation partner and "Explain My Answer" grammar explanations
- ☐ Adaptive difficulty based on learner mistakes and weak skills
- ☐ Placement check for new users to start at their actual level
- ☐ Community corrections (native-speaker review of written/spoken submissions)
- ☐ Cultural insights across the francophone world at every level
- ☐ Responsive design works on mobile and desktop
- ☐ Audio playback for listening exercises
- ☐ Timer functionality for tests
- ☐ Built with Next.js composable components (Server + Client Components where appropriate)
- ☐ Content mapped to the Complete French All-in-One book